

Name:- Moreshwar Haldnkar Roll No:- CE11

Task 1: Import NumPy, Pandas, Matplotlib.pyplot and Seaborn.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Task 2: Load train.csv into a Pandas DataFrame using pd.read\_csv().

```
from google.colab import files
uploaded = files.upload()
df = pd.read_csv('Titanic-Dataset (1).csv')
df.head()
```

Task 3: Display first 10 rows using df.head(10).

```
print("First 10 Rows:")
print(df.head(10))
```

Task 4: Display last 5 rows using df.tail(5)

```
print("\nLast 5 Rows:")
print(df.tail(5))
```

Task 5 :Check dataset dimensions using df.shape (Expected: 891 x 12).

```
print("\nDataset Shape:", df.shape)
```

Task 6:Display dataset information using df.info().

```
print("\nDataset Info:")
print(df.info())
```

Task 7: Generate summary statistics using df.describe(); record mean, minimum and maximum values of Age and Fare

```
print("\nSummary Statistics:")
print(df.describe())

print("\nAge Mean:", df['Age'].mean())
print("Age Min:", df['Age'].min())
print("Age Max:", df['Age'].max())

print("\nFare Mean:", df['Fare'].mean())
print("Fare Min:", df['Fare'].min())
print("Fare Max:", df['Fare'].max())
```

Task 8: Identify missing values using df.isnull().sum().

```
print("\nMissing Values:")
print(df.isnull().sum())
```

Task 9: Fill missing Age values with the median

```
print("\nMissing Values After Filling Age:")
print(df.isnull().sum())
```

Task 10: Filter surviving passengers (Survived == 1).

```
survivors = df[df['Survived'] == 1]
print("\nSurvivors Data:")
print(survivors.head())
```

Task 11: Convert Fare column to a NumPy array.

```
fare_array = df['Fare'].to_numpy()
```

```
mean_fare = np.mean(fare_array)
print(f"Mean Fare: {mean_fare}")
```

Task 12: Calculate maximum and average Fare using np.max() and np.mean().

```
print("\nMax Fare:", np.max(fare_array))
print("Average Fare:", np.mean(fare_array))
```

Task 13: Plot histogram of Age.

```
plt.figure()
plt.hist(df['Age'])
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Frequency")
plt.show()
```

Task 14: Plot bar chart of Survival Count.

```
plt.figure()
df['Survived'].value_counts().plot(kind='bar')
plt.title("Survival Count")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Count")
plt.show()
```

Task 15: Plot scatter graph of Age vs Fare.

```
plt.figure()
plt.scatter(df['Age'], df['Fare'])
plt.title("Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```

Name:- Maruti N. Patil Roll No:- CE35

Task 1: Import NumPy, Pandas, Matplotlib.pyplot and Seaborn.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Task 2: Load train.csv into a Pandas DataFrame using pd.read\_csv().

```
from google.colab import files
uploaded = files.upload()
df = pd.read_csv('Titanic-Dataset (1).csv')
df.head()
```

No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving Titanic-Dataset (1).csv to Titanic-Dataset (1) (2).csv

|   | PassengerId | Survived | Pclass | Name                                              | Sex    | Age  | SibSp | Parch | Ticket           | Fare    | Cabin | Embarked |
|---|-------------|----------|--------|---------------------------------------------------|--------|------|-------|-------|------------------|---------|-------|----------|
| 0 | 1           | 0        | 3      | Braund, Mr. Owen Harris                           | male   | 22.0 | 1     | 0     | A/5 21171        | 7.2500  | NaN   | S        |
| 1 | 2           | 1        | 1      | Cumings, Mrs. John Bradley (Florence Briggs Th... | female | 38.0 | 1     | 0     | PC 17599         | 71.2833 | C85   | C        |
| 2 | 3           | 1        | 3      | Heikkinen, Miss. Laina                            | female | 26.0 | 0     | 0     | STON/O2. 3101282 | 7.9250  | NaN   | S        |

Task 3: Display first 10 rows using df.head(10).

```
print("First 10 Rows:")
print(df.head(10))
```

```
First 10 Rows:
  PassengerId  Survived  Pclass \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3
5            6         0       3
6            7         0       1
7            8         0       3
8            9         1       3
9           10         1       2

   Name                               Sex  Age  SibSp  \
0  Braund, Mr. Owen Harris             male  22.0    1
1  Cumings, Mrs. John Bradley (Florence Briggs Th... female  38.0    1
2                Heikkinen, Miss. Laina      female  26.0    0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0    1
4            Allen, Mr. William Henry       male  35.0    0
5            Moran, Mr. James              male   NaN    0
6      McCarthy, Mr. Timothy J             male  54.0    0
7      Palsson, Master. Gosta Leonard       male   2.0    3
8  Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg) female  27.0    0
9      Nasser, Mrs. Nicholas (Adele Achem)  female  14.0    1

   Parch  Ticket   Fare Cabin Embarked
0      0   A/5 21171   7.2500   NaN      S
1      0    PC 17599  71.2833   C85      C
2      0  STON/O2. 3101282   7.9250   NaN      S
3      0   113803  53.1000  C123      S
4      0   373450   8.0500   NaN      S
5      0   330877   8.4583   NaN      Q
6      0   17463   51.8625  E46      S
7      1   349909  21.0750   NaN      S
8      2   347742  11.1333   NaN      S
9      0   237736  30.0708   NaN      C
```

Task 4: Display last 5 rows using df.tail(5)

```
print("\nLast 5 Rows:")
print(df.tail(5))
```

```
Last 5 Rows:
  PassengerId  Survived  Pclass                               Name \
886          887         0       2                Montvila, Rev. Juozas
887          888         1       1      Graham, Miss. Margaret Edith
888          889         0       3  Johnston, Miss. Catherine Helen "Carrie"
889          890         1       1      Behr, Mr. Karl Howell
890          891         0       3      Dooley, Mr. Patrick

   Sex  Age  SibSp  Parch  Ticket   Fare Cabin Embarked
886  male  27.0    0      0   211536  13.00   NaN      S
887  female  19.0    0      0   112053  30.00  B42      S
888  female   NaN    1      2  W./C. 6607  23.45   NaN      S
889  male  26.0    0      0   111369  30.00  C148      C
890  male  32.0    0      0   370376   7.75   NaN      Q
```

Task 5 :Check dataset dimensions using df.shape (Expected: 891 x 12).

```
print("\nDataset Shape:", df.shape)
```

Dataset Shape: (891, 12)

Task 6: Display dataset information using df.info().

```
print("\nDataset Info:")
print(df.info())
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   PassengerId            891 non-null    int64
1   Survived               891 non-null    int64
2   Pclass                 891 non-null    int64
3   Name                   891 non-null    object
4   Sex                    891 non-null    object
5   Age                    714 non-null    float64
6   SibSp                  891 non-null    int64
7   Parch                  891 non-null    int64
8   Ticket                 891 non-null    object
9   Fare                   891 non-null    float64
10  Cabin                  204 non-null    object
11  Embarked               889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None
```

Task 7: Generate summary statistics using df.describe(); record mean, minimum and maximum values of Age and Fare

```
print("\nSummary Statistics:")
print(df.describe())

print("\nAge Mean:", df['Age'].mean())
print("Age Min:", df['Age'].min())
print("Age Max:", df['Age'].max())

print("\nFare Mean:", df['Fare'].mean())
print("Fare Min:", df['Fare'].min())
print("Fare Max:", df['Fare'].max())
```

```
Summary Statistics:
      PassengerId  Survived  Pclass    Age    SibSp  \
count    891.000000   891.000000   891.000000  714.000000  891.000000
mean         446.000000     0.383838     2.308642   29.699118    0.523008
std         257.353842     0.486592     0.836071   14.526497    1.102743
min           1.000000     0.000000     1.000000     0.420000    0.000000
25%         223.500000     0.000000     2.000000   20.125000    0.000000
50%         446.000000     0.000000     3.000000   28.000000    0.000000
75%         668.500000     1.000000     3.000000   38.000000    1.000000
max         891.000000     1.000000     3.000000   80.000000    8.000000

      Parch    Fare
count    891.000000  891.000000
mean         0.381594   32.204208
std         0.806057   49.693429
min          0.000000    0.000000
25%          0.000000    7.910400
50%          0.000000   14.454200
75%          0.000000   31.000000
max          6.000000  512.329200

Age Mean: 29.69911764705882
Age Min: 0.42
Age Max: 80.0

Fare Mean: 32.204207968574636
Fare Min: 0.0
Fare Max: 512.3292
```

Task 8: Identify missing values using df.isnull().sum().

```
print("\nMissing Values:")
print(df.isnull().sum())
```

```
Missing Values:
PassengerId    0
```

```

Survived      0
Pclass        0
Name          0
Sex           0
Age          177
SibSp         0
Parch         0
Ticket        0
Fare          0
Cabin        687
Embarked      2
dtype: int64

```

Task 9: Fill missing Age values with the median

```

print("\nMissing Values After Filling Age:")
print(df.isnull().sum())

```

```

Missing Values After Filling Age:
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            0
SibSp           0
Parch           0
Ticket          0
Fare            0
Cabin          687
Embarked        2
dtype: int64

```

Task 10: Filter surviving passengers (Survived == 1).

```

survivors = df[df['Survived'] == 1]
print("\nSurvivors Data:")
print(survivors.head())

```

```

Survivors Data:
  PassengerId  Survived  Pclass  \
1             2         1       1
2             3         1       3
3             4         1       1
8             9         1       3
9            10         1       2

   Name                               Sex  Age  SibSp  \
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0    1
2                               Heikkinen, Miss. Laina  female  26.0    0
3  Futrelle, Mrs. Jacques Heath (Lily May Peel)      female  35.0    1
8  Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)  female  27.0    0
9    Nasser, Mrs. Nicholas (Adele Achem)              female  14.0    1

   Parch  Ticket   Fare Cabin Embarked
1      0   PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250  NaN        S
3      0   113803   53.1000  C123        S
8      2   347742  11.1333  NaN        S
9      0   237736  30.0708  NaN        C

```

Task 11: Convert Fare column to a NumPy array.

```
fare_array = df['Fare'].to_numpy()
```

```

mean_fare = np.mean(fare_array)
print(f"Mean Fare: {mean_fare}")

```

```
Mean Fare: 32.204207968574636
```

Task 12: Calculate maximum and average Fare using np.max() and np.mean().

```

print("\nMax Fare:", np.max(fare_array))
print("Average Fare:", np.mean(fare_array))

```

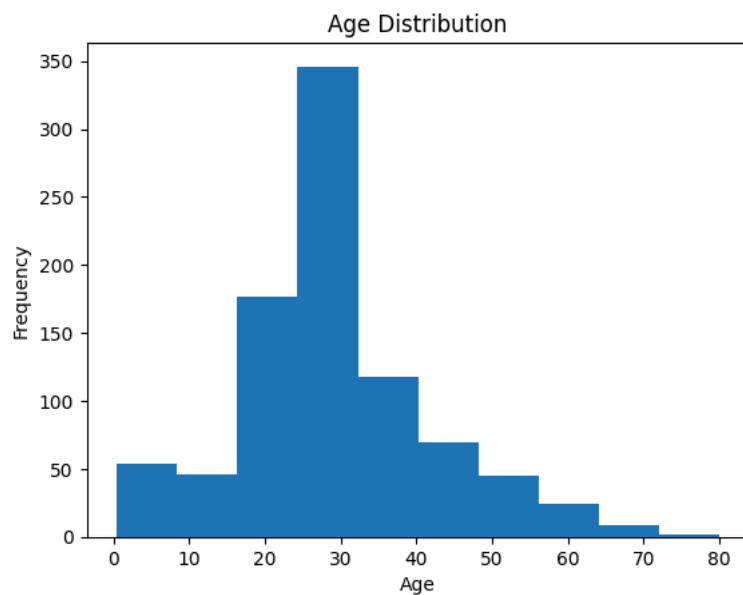
```

Max Fare: 512.3292
Average Fare: 32.204207968574636

```

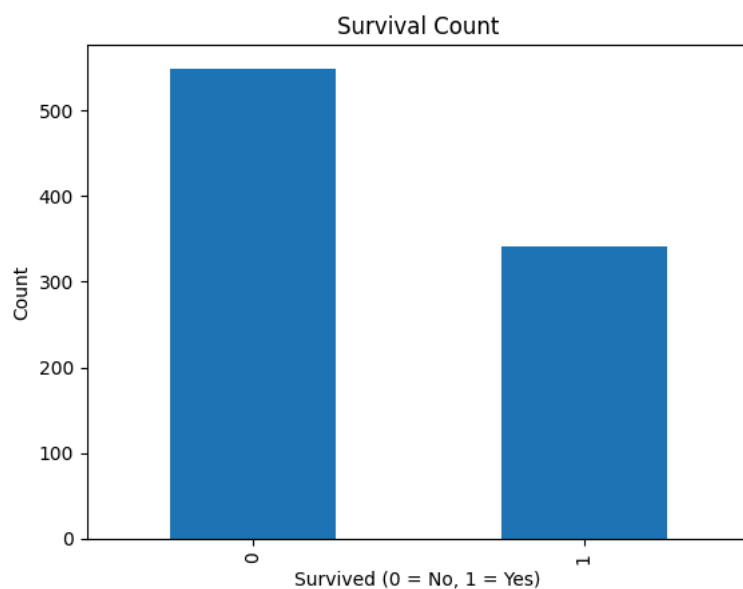
Task 13: Plot histogram of Age.

```
plt.figure()
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plt.ylabel("Frequency")
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Task 14: Plot bar chart of Survival Count.

```
plt.figure()
df['Survived'].value_counts().plot(kind='bar')
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plt.ylabel("Count")
plt.show()
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Task 15: Plot scatter graph of Age vs Fare.

```
plt.figure()
plt.scatter(df['Age'], df['Fare'])
plt.title("Age vs Fare")
plt.xlabel("Age")
plt.ylabel("Fare")
plt.show()
```

